

FAQ and Troubleshooting

This document contains frequently asked questions and troubleshooting guidance for EGC systems, including firmware upgrade procedures.

Firmware Upgrade

The firmware upgrade for the Microgrid Controller, i.e., software version upgrade, is typically performed by the operations and maintenance (O&M) team. You must first obtain the Microgrid Controller firmware package corresponding to the specific hardware, then select the appropriate upgrade method based on whether it is connected to the cloud:

Connected to EnOS Cloud: Use Logger OTA for remote and batch upgrades.

Not Connected to EnOS Cloud: Use the Logger Local O&M Tool for local, single-device upgrades.

OTA remote upgrade

Step 1 (Operated by O&M team): Obtain the Microgrid Controller firmware package and release notes file.

Step 2 (Operated by O&M team): Navigate to Cloud Asset Configuration Center > Edge Management > Logger OTA menu, go to the Firmware Management tab, and click New Firmware.

Step 3 (Operated by O&M team): Select the product category, upload the corresponding release notes file and firmware package, and click Confirm.

Step 4: Navigate to the OTA Jobs tab and click New Job.

Step 5: Select the new firmware version to be upgraded. The system will automatically filter out Edge Logger devices that can be upgraded to the specified version. Choose the devices to be upgraded based on the actual situation. Enter the task name and upgrade start time, then click Confirm Submission at the bottom of the page. Note: If the new firmware version has never been used for an upgrade in the current environment before, the backend will restrict the initial upgrade to a single Logger. After a successful single-device upgrade, the firmware version can be used for batch upgrades.

Step 6: After submission, you can view the current upgrade tasks in the task list. Navigate to the upgrade details and refresh the page to check the current task status, progress, and results.

Step 7: Navigate to the Upgrade Record tab to view all upgrade records for the current device.

10.2 Local upgrade

Step 1: Obtain the Microgrid Controller firmware package. Step 2: Navigate to Local O&M Tool > O&M Management > Firmware Upgrade, click Firmware Upgrade, upload the firmware package, and wait for a few minutes. Step 3: Upon completion, the system will automatically restart. After restarting, revisit the Local O&M Tool to check whether the current firmware version has been updated.

Frequently Asked Questions

How to Change the Cloud Environment for the Current Device

Cloud Version 2.3.0 and Above

According to the two registration and activation methods of the current device, the steps to change the cloud environment are different, including the following two:

Edge Installer App Registration & Edge Activation Method

Before changing the cloud, you need to delete the registered assets on the Edge Installer App. The procedure is as follows.

Step 1: Power on the device.

Step 2: Log in to the Edge Installer App, find the Edge Logger gateway device to change the environment, click on the upper right corner, and select "Delete" to delete devices. You can select all devices to delete.

Precautions:

- This step is only required when connecting devices.

Step 3: Click on the upper right corner again and select "Delete". Make a selection to delete Edge and confirm deletion.

Step 4: For the device, you can continue to refer to the Edge Registration and Activation section, and select other clouds for registration and activation.

Precautions:

- To change the cloud and re-connect the Edge devices by the Edge Installer App, the original registration information must be deleted first.

- Connecting devices by the Edge Installer App does not support simultaneous access to two clouds

Cloud Asset Configuration Registration & Edge Activation Method

Before changing the cloud, you need to delete the registered assets in the device. The procedure is as follows.

Step 1: Power on the device and log in to the local Edge Logger page.

Step 2: Enter the device registration page, select "Clear Registration" and confirm.

Step 3: For the device, you can continue to refer to the Edge Registration and Activation section, and select different clouds for registration and activation.

Precautions:

- This step only supports cloud version 2.3.0 and above.
- If the cloud version is lower than 2.3.0, refer to How to connect to the cloud environment (version 2.2) .

Cloud Version 2.2

You must first obtain all domain name configuration items in the cloud, and write the configuration to the device through the local page. The specific steps are as follows:

Step 1: Contact the Edge after-sales O&M team to obtain cloud configuration items. The specific configuration items are as follows.

Description

Configuration ID

Example (PPE Cloud)

Notes

Cloud-related Environment Configuration - Required

Cloud MQTT Broker address

edgelogger.client.mqtt.broker.url

tcp://mqtt-ppe1.envisioniot.com:21883

ssl://mqtt-ppe1.envisioniot.com:18883

For encryption: ssl

No encryption: tcp

Alternative

Cloud HTTP Broker address

edgelogger.ota.http.broker.url

https://iot-http-ppe1.envisioniot.com

Required

Cloud API gateway address

edgelogger.client.api.server.url

<https://apim-ppe1.envisioniot.com>

Required

Cloud API application registration information

edgelogger.client.api.accessKey

f13a8249-562a-43b9de6b99d8-acfb-42fc

Required

edgelogger.client.api.secretKey

de6b99d8-acfb-42fc-980f-08d82ca6966e

Required

Cloud release resource package download domain name

edgelogger.client.resource.download.domain

<https://portal-ppe1.envisioniot.com>

Required

Message integration channel in the cloud

edgelogger.client.http.integration.channel

<https://iot-http-integration-ppe1.envisioniot.com>

Required

Monitoring Environment Configuration - Configuration On Demand

O&M monitoring pushGateway address

edge.operation.pushgateway.address

<https://pushgateway-ppe1.envisioniot.com>

Configure as needed

O&M monitoring pushGateway user/password

edge.operation.pushgateway.user.name

pushgateway

Configure as needed

edge.operation.pushgateway.user.password

envision_root

Configure as needed

O&M Tunnel Configuration - Configure On Demand

rttys server nginx address or IP

edge.rtty.rttys.url

edge-ops-ppe1.envisioniot.com

Configure as needed

rttys server port number

edge.rtty.rttys.port

32751

Configure as needed

rttys certificate server nginx address

edge.rtty.cert.server.url

https://edge-cloudcore-ppe1.envisioniot.com:10002

Configure as needed

Step 2: Log in to the local page of Edge Logger.

Step 3: Enter the "O&M Management -> Function" page, modify the advanced configuration items, and click .

Step 4: Modify the configuration file lion.properties, and modify them one by one according to the environment configuration identifiers in the table in step 1. Finally, click "Save".

Precautions:

- # indicates comments. By default, the configuration file is in the comment status, which needs to be deleted in step 4.

- Special characters such as spaces are not allowed before and after =.

How to Replace Asset Information to Other Physical Devices

Prerequisite: This chapter is applicable only to devices registered by the "Cloud Asset Configuration Tool Registration & Edge Activation" method. It is not applicable to devices registered by the "Edge Installer App Registration & Edge Activation" method.

Assumption: Device A has hardware damage (for example: serial port damage) during use, and you want to return it to the factory for repair. However, at this time, the configuration information for cloud access (for example: connection, device mounting and templates, etc.) has been basically completed. Access can continue using the standby device B.

Step 1: Power on device A and log in to the local page of Edge Logger.

Step 2: Enter the "O&M Management -> System Management" menu, click "Restore Default Configuration", read the contents of the pop-up box to restore the default configuration of the device, and then click "OK".

Step 3: Shut down the system of device A. Click "System Shutdown", and click "Shutdown" in the confirmation box. At this time, device A is waiting to be returned to the factory for repair.

Step 4: Log in to the cloud asset configuration and download the box.conf of device A. Standby device B can be powered on. Perform the replacement process in accordance with the Cloud Asset Configuration Registration & Edge Activation section.

Precautions:

- In step 4, the registration and activation process of standby device B directly uses device A's box.conf.
- If device A is connected and registered by the Edge Installer App Registration & Edge Activation method, then this method is not applicable. For the standby device B, you need to configure connection and device information. (The configuration information of device A in the cloud remains unchanged. Device A can continue to be used when it returns to the site after repair.)

How to Restore Default Configuration

According to the two registration and activation methods of the current device, the steps to restore the default configuration are different, including the following two:

Case 1: Use the Edge Installer App Registration & Edge Activation Method

Assets that have been registered on the Edge Installer App need to be deleted. The procedure is as follows.

Step 1: Power on the device.

Step 2: Log in to the Edge Installer App, find the Edge Logger gateway device to change the environment, click on the upper right corner, and select "Delete" to delete devices. You can select all devices to delete.

Precautions:

- This step is only required when connecting devices.

Step 3: Click on the upper right corner again and select "Delete". Make a selection to delete Edge and confirm deletion. Wait for about 3 minutes before proceeding to the next step.

Step 4: Log in to the local page of Edge Logger, enter the "O&M Management -> System Management" page and click "System Shutdown". A dialog box will pop up to confirm the shutdown.

Precautions:

- To change the cloud and re-connect the Edge devices by the Edge Installer App, the original registration information must be deleted first.
- Connecting devices by the Edge Installer App does not support registering two clouds at the same time.

Case 2: Use the Cloud Asset Configuration Registration & Edge Activation Method

Step 1: Power on the device and log in to the local Edge Logger page.

Step 2: Enter the "O&M Management -> System Management" menu, click "Restore Default Configuration", read the contents of the pop-up box to restore the default configuration of the device, and then click "OK".

Step 3: Shut down the system of device A. Click "System Shutdown", and click "Shutdown" in the confirmation box. At this point, device A has returned to the default status.

How to Check the Points Collected and Sent to Cloud

Viewing by Cloud Asset Configuration Tool

Steps: Enter the device asset page, select the Edge gateway device, click the "Measurement Point" page, you can view the corresponding values of the two fields. The meanings are as follows.

receive_point_count: Collected points/minute

send_point_count: Points sent to cloud/minute

Viewing by Local O&M Tool

Steps: Enter the overview and select the Gateway Status page, and view the corresponding value through the running status bar.

How to View Cloud Communication Problems

When the device cannot connect to the cloud, refer to the following steps:

Step 1: Log in to the local page of Edge Logger.

Step 2: Enter the "Network Configuration -> Other" page, and then click in the "Network Connectivity" column.

Step 2: Ping baidu.com and test DNS one by one. Normally, the feedback should be connected, which indicates the device can access the Internet.

Precautions

If baidu.com cannot be reached, it indicates that the device cannot access the Internet, and you need to check how the device accesses the Internet

If the DNS cannot be connected, it indicates that there is a problem with the automatically assigned DNS. You need to contact the operator to access the Internet with the SIM card of the device. In the case of a wired connection, you need to contact the network infrastructure to view the DHCP server.

Step 3: Enter the "O&M Management -> System Diagnosis" page, and then click "Start Diagnosis". Observe the network connectivity check bar.

Precautions

- The abnormality logs in the log check items on the self-test page are for reference only, and are not used as guidelines for O&M.
- The log content is time-sensitive, and it will be repeated downloaded.

How to View Publishing Issues

Step 1: Enter the basic information page of the Edge management, check whether the connection status to the cloud is connected. If disconnected, jump to Check cloud communication problems to troubleshoot, otherwise proceed to the next step.

Step 2: Click the "Publish" button.

Step 3: Enter the device asset page, select the Edge gateway device, select the "Command" bar, and check whether there is an update_resource field in responded status. If yes, proceed to next step; otherwise, contact edge-support@envision-digital.com.

Step 4: Enter the device asset page, select the Edge gateway device, select the "Measurement Point" column, and check the value corresponding to update_result. The time is close to the time of clicking "Publish" in step 2. If the time is refreshed and the value is 1, indicating that the actual device has been published successfully but there is a problem with the platform environment. At this time, the device access is unaffected, and the access can continue.

Precautions:

- If it is a new OU, refer to the OU application and initialization chapter in the Edge Delivery Operations Guide.

How to Upgrade the Edge Logger Firmware

Overview: Edge Logger firmware upgrade, that is, software version upgrade, is usually operated by the O&M team. You need to first obtain the Edge Logger firmware package corresponding to the specific hardware type, and then select the corresponding upgrade method according to whether Logger is connected to the cloud.

- Logger connected to EnOS Cloud: OTA can be used for remote and batch upgrade.

- Logger not connected to EnOS Cloud: You can use the local O&M tool of Logger for local and single upgrade.

OTA Remote Upgrade

Step 1 (operated by O&M personnel): Obtain the Edge Logger firmware package and release notes file.

Step 2 (operated by O&M personnel): Enter the "Cloud Asset Configuration Center > Edge Management > Logger OTA" menu, enter the "Firmware Management" tab page, and click "New Firmware".

Step 3 (operated by O&M personnel): Select the product category, upload the corresponding release notes file and firmware package, and click "Confirm".

Step 4: Enter the "OTA Jobs" tab page and click "New Job".

Step 5: Select the new firmware version that needs to be upgraded, and the system will automatically screen out Loggers that can be upgraded to the specified version. Select the device that needs to be upgraded according to the actual situation. Fill in the task name and upgrade start time and click "Confirm Submission" at the bottom of the page after completion.

Note: If the new version of the firmware package has never been upgraded in the current environment before, the background will limit that only one Logger can be selected for the first upgrade. After a single unit is successfully upgraded, the firmware package of this version can be used for batch upgrade.

Step 6: After submitting, you can view the current upgrade tasks in the task list. Enter the upgrade details and refresh the page to view the current task status, progress, and results.

Step 7: You can enter the "Upgrade Record" tab page to view all the upgrade records of the current device.

Local Upgrade

Step 1: Get the Edge Logger firmware package.

Step 2: Enter the "Local O&M Tool > O&M Management > Firmware Upgrade" menu, click "Firmware Upgrade", upload the firmware package, and wait for a few minutes.

Step 3: Once completed, the system will automatically reboot. After rebooting, you can revisit the local O&M tool to check whether the current firmware version has been updated.

Logger Port Number Usage

Port No.

Usage

Content

Remark

22

SSH

Open for external

-

53

DNS Server

Only open during wifi configuration

Only for DTU and HPU Lite

67

Assign IP to devices accessing AP

Only open during wifi configuration

Only for DTU and HPU Lite

80

Nginx Proxy

Open for external

-

External Terminal Resistor Installation Instruction

Taking DTU and SIEMENS 6SL3255-0VC00-0HA0 as examples, the termination resistor installation steps are as follows:

Siemens RS485 terminal resistor USS/Modbus RTU communication
Siemens RS485 terminal resistor
USS/Modbus RTU communication

Siemens RS485 terminal resistor USS/Modbus RTU communication

Siemens RS485 terminal resistor USS/Modbus RTU communication

Step1: Connect the blue P+ cable to RS485 bus A and the blue N- cable to RS485 bus B.

Step 2: Wrap the terminals of the two white wires with insulating tape.